

Generation Interconnection Request System Impact Study Report

For Queue Project Q20-1435

Curecanti 345 kV
July 24, 2021

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q20-1435. The Interconnection Customer proposes a 341.2 MW solar plus storage facility interconnecting at the Curecanti 345 kV substation in the Pennsylvania area. The request was received on March 20, 2020. The study evaluated system intact and single contingency conditions using the regional planning case, including 482 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 1 overloaded facility. The total estimated cost of the network upgrades allocated to this project is \$4,342,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

1. Facility: Curecanti 138-1 kV Auto
Contingency: Curecanti – Moser Generating Sta 345 kV L
Loading: 109.9% (Rating: 390 MVA)
DFAX: 13.7%
Pre-project loading: 97.9%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Curecanti 138-1 kV Autotransformer	Replace transformer with higher-rated unit	\$7,104,000	61% (\$4,342,000)
	Total allocated to project		\$4,342,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.